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Inventor(s)	Lee; Sanggoo et al.

Display apparatus

Abstract

A display apparatus is provided. The display apparatus includes: a display panel; a support bracket coupled to a rear surface of the display panel and defining a locking groove; a support arm coupled to the support bracket and supporting the display panel, wherein the support arm is configured to rotate between a first and second orientations; and a support stand supporting the support arm and configured to allow the support arm to move along a vertical direction between first and second positions. The support arm includes: a rotating link configured to rotate as the support arm moves; and a locking link configured to move from a locked position in which the locking link is inserted into the locking groove to restrict rotation of the display panel, to an unlocked position in which the locking link is withdrawn from the locking groove to allow rotation of the display panel.

Inventors: **Lee; Sanggoo (Suwon-si, KR), Kim; Sunggi (Suwon-si, KR), Park; Daesik (Suwon-si, KR), Song; Hyewon (Suwon-si, KR), Yun; Hyunwoong (Suwon-si, KR), Lee; Jeongroh (Suwon-si, KR), In; Woosung (Suwon-si, KR)**

Applicant: **SAMSUNG ELECTRONICS CO., LTD.** (Suwon-si, KR)

Family ID: **88307707**

Assignee: **SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)**

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Primary Examiner: Parker; Allen L

Assistant Examiner: Crum; Gage

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This is a continuation of International Application No. PCT/KR2023/000872, filed on Jan. 18, 2023, which is based on and claims priority to Korean Patent Application No. 10-2022-0048029, filed on Apr. 19, 2022, in the Korean Intellectual Property Office, and Korean Patent Application No. 10-2022-0060877, filed on May 18, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND

1. Field

(1) The disclosure relates to a display apparatus including a display module and a support device for supporting the display module to enable a rotation and an upward and downward movement of the display module.

2. Description of Related Art

(2) A display apparatus is an output apparatus for converting acquired or stored electrical information into visual information and displaying the converted electrical information to a user.

(3) The display apparatus may include a display module for displaying an image and a support device for supporting the display module. The support device may be provided to support the display module so that a front surface of the display module faces the user.

(4) The support device may support the display module to be movable upward and downward within a predetermined range. In addition, the support device may support the display module to perform a pivot rotation.

(5) When the display module is rotated without being sufficiently spaced apart from a floor surface, the display module may collide with the floor surface and thus the display module may be damaged.

SUMMARY

(6) Provided is a display apparatus may prevent a display module from colliding with a floor surface due to being rotated.

(7) Further, provided is a display apparatus includes a display module which is able to rotate with respect to a support device only when the display module is spaced apart from the floor surface by a predetermined distance or more.

(8) Further still, provided is a display apparatus restricts rotation of a display module at a position having a possibility of colliding with a surface if rotated.

(9) The technical objectives of the disclosure are not limited to the above, and other objectives may become apparent to those of ordinary skill in the art based on the following descriptions.

(10) In accordance with an aspect of the disclosure, a display apparatus includes: a display panel; a support bracket coupled to a rear surface of the display panel and defining a locking groove; a support arm coupled to the support bracket and supporting the display panel, wherein the support arm is configured to rotate between a first orientation in which a long side of the display panel is laterally disposed and a second orientation in which the long side of the display panel is longitudinally disposed; and a support stand supporting the support arm and configured to allow the support arm to move along a vertical direction between a first position and a second position lower than the first position. The support arm includes: a rotating link configured to rotate in a first direction as the support arm moves from the second position to the first position; and a locking link configured to move from a locked position in which the locking link is inserted into the locking groove to restrict rotation of the display panel, to an unlocked position in which the locking link is withdrawn from the locking groove to allow rotation of the display panel, based on the rotating link rotating in the first direction.

(11) The support arm may further include a fixing link coupled to the support stand.

(12) The locking link may be configured to laterally move along a horizontal direction with respect to the fixing link, and the locking link may be inserted into the locking groove by moving toward

the support bracket, and may be withdrawn from the locking groove by moving away from the support bracket.
(13) The support arm may further include an elastic member configured to bias the locking link toward the support bracket.
(14) One end of the elastic member may be connected to the fixing link, and another end of the elastic member may be connected to the locking link.
(15) The locking link may be configured to move to the locked position based on an elastic force of the elastic member when the support arm moves from the first position to the second position.
(16) The rotating link may be configured to rotate in a second direction opposite to the first direction based on the locking link moving from the unlocked position to the locked position.
(17) The locking link may include: a locking portion configured to extend be inserted into the locking groove or withdrawn from the locking groove; and an extension portion extending toward the support stand and defining an insertion groove.
(18) The rotating link may include: a first protrusion that is provided in into the insertion groove; and a second protrusion that extends into the support stand and is configured to rotate together with the first protrusion.
(19) The support stand may include a rotation guide portion configured to contact the second protrusion and rotate the rotating link in the first direction as the support arm moves from the second position to the first position.
(20) The rotation guide portion may be provided adjacent to an upper end of the support stand.
(21) The rotation guide portion may include a rotation guide surface provided at an angle oblique to a horizontal direction, and the rotation guide surface may be configured to guide the second protrusion such that the second protrusion rotates in the first direction to be arranged in parallel with the rotation guide surface.
(22) The support bracket may be detachably coupled to the rear surface of the display panel.
(23) In accordance with an aspect of the disclosure, a display stand includes: a support bracket having a locking groove formed therein, and configured to support a display panel and rotate between a first orientation and a second orientation; a support arm configured to support the support bracket; and a support stand configured to slidably support the support arm between a first position and a second position lower than the first position. The support arm includes: a rotating link configured to rotate in a first direction as the support arm moves from the second position to the first position; and a locking link configured to move from a locked position in which the locking link is inserted into the locking groove to restrict rotation of the display panel, to an unlocked position in which the locking link is withdrawn from the locking groove to allow rotation of the display panel, based on the rotating link rotating in the first direction.
(24) The support arm may be configured to move in a vertical direction between the first position and the second position, and the locking link may be configured to move along a horizontal direction that is perpendicular to the vertical direction between the locked position and the unlocked position.
(25) The locking groove may be aligned with the locking link based on the support bracket being in the first orientation, and another locking groove may be formed in the support bracket and may be aligned with the locking link based on the support bracket being in the second orientation.
(26) The support arm may further include a fixed portion, and the locking link may be configured to slide along the fixed portion in the horizontal direction.
(27) The display stand may further include a spring connected between the fixed portion and the locking link, and configured to bias the locking link to the locked position.
(28) In accordance with an aspect of the disclosure, a method of limiting rotation of a support bracket that is slidable along a support stand between a first position and a second position lower than the first position, includes: restricting rotation of the support bracket based on the support bracket being at the second position; sliding the support bracket from the second position to the

first position; and withdrawing a locking link from a locking groove formed in the support bracket based on the support bracket sliding from to the first position.
(29) A rotating link may extend into a groove formed in the locking link, and the method may further include rotating the rotating link in a first direction as the support bracket moves to the first position to move the locking link.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description, taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a perspective view illustrating a display apparatus according to an embodiment;
(3) FIG. **2** is a rear perspective view illustrating the display apparatus shown in FIG. **1** according to an embodiment;
(4) FIG. **3** is a perspective view illustrating a display apparatus according to an embodiment;
(5) FIG. **4** is a front exploded perspective view illustrating a display apparatus according to an embodiment;
(6) FIG. **5** is a rear exploded perspective view illustrating a display apparatus according to an embodiment;
(7) FIG. **6** is a view illustrating a support bracket of a display apparatus according to an embodiment;
(8) FIG. **7** is a view illustrating the support bracket shown in FIG. **6** according to an embodiment;
(9) FIG. **8** is a view illustrating an internal structure of a support arm when a pivot rotation of a display module is restricted in a display apparatus according to an embodiment;
(10) FIG. **9** is a view illustrating an internal structure of a support arm when a pivot rotation of a display module is allowed in a display apparatus according to an embodiment;
(11) FIG. **10** is a cross-sectional view taken along line A-A′ of FIG. **8** according to an embodiment; and
(12) FIG. **11** is a cross-sectional view taken along line B-B′ of FIG. **9** according to an embodiment.
DETAILED DESCRIPTION
(13) Embodiments described in the specification and configurations shown in the accompanying drawings are provided as examples, and various modifications may replace the embodiments and the drawings of the disclosure at the time of filing of the application.
(14) Further, identical symbols or numbers in the drawings of the disclosure denote components or elements configured to perform substantially identical functions.
(15) Further, terms used herein are only for the purpose of describing particular embodiments and are not intended to limit to the disclosure. The singular form is intended to include the plural form as well, unless the context clearly indicates otherwise. It should be further understood that the terms “include,” “including,” “have,” and/or “having” specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
(16) Further, it should be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, the elements are not restricted by the terms, and the terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and similarly, a second element could be termed a first element without departing from the scope of the disclosure. The term “and/or” includes combinations of one or more or all of a plurality of associated listed items.

(17) The terms “front”, “rear”, “back”, “left”, “right”, and the like as used herein are defined with respect to the drawings, but the terms may not restrict the shape and position of the respective components.
(18) Hereinafter, embodiments will be described in detail with reference to the accompanying drawings.
(19) FIG. **1** is a perspective view illustrating a display apparatus according to an embodiment. As shown in FIG. **1**, a display module may be laterally disposed. FIG. **2** is a rear perspective view illustrating the display apparatus shown in FIG. **1**. FIG. **3** is a perspective view illustrating a display apparatus according to an embodiment, which shows a state in which a display module is longitudinally disposed.
(20) A display module **10** is a device capable of processing an image signal and visually displaying a processed image. The processed image may include information, reference materials, data, and the like as characters, figures, graphs, images, and the like. The display module **10** may include a television, a monitor, and the like.
(21) The display module **10** may be configured to display a screen. The display module **10** may include a self-emissive display panel, such as an organic light-emitting diode (OLED), or a non-emissive display panel, such as a liquid crystal display (LCD). There is no particular limitation on the type of the display panel.
(22) The display module **10** may be provided so that the length of a lateral side and the length of a longitudinal side are different from each other. That is, the display module **10** may be provided to have a long side **11** and a short side **12**. The display module **10** may be provided in a rectangular plate shape. In addition, as shown in FIGS. **1** to **3**, the display module **10** may be provided as a curved display in which the long side **11** is formed in a rounded shape. Alternatively, the display module **10** may be provided as a flat display in which the long side **11** is provided as a straight line.
(23) According to an embodiment, the display apparatus **1** may include a display module **10** and support devices **20**, **100**, and **30** supporting the display module **10**.
(24) The support devices **20**, **100**, and **30** may include a support bracket **20**, a support arm **100**, and a support stand **30**.
(25) Referring to FIG. **2**, the support bracket **20** may be coupled to a rear side of the display module **10**. The support arm **100** may be coupled to the support bracket **20** to support the display module **10**. The support arm **100** may support the display module **10** such that the display module **10** may pivot. The support arm **100** may be coupled to the support stand **30** to be movable in an up direction and a down direction (i.e., along a vertical direction) within a predetermined range.
(26) Referring to FIGS. **1** and **3**, the display module **10** may be positioned in any one of a first orientation in which the long side **11** is laterally disposed and a second orientation in which the long side **11** is longitudinally disposed. The display module **10** may be positioned in the second orientation by pivot-rotating from the first orientation, and may be positioned in the first orientation by pivot-rotating from the second orientation. Hereinafter, “pivot rotation” may be expressed as “rotation”.
(27) The support arm **100** may support the support bracket **20** and the display module **10** coupled to the support bracket **20** for rotation between the first orientation and the second orientation. The first orientation may be referred to as a landscape orientation, and the second orientation may be referred to as a portrait orientation.
(28) The support arm **100** may be coupled to the support stand **30** to be movable upward and downward from a lowermost end of the support stand **30** to an uppermost end of the support stand **30**. As the support arm **100** moves upward and downward with respect to the support stand **30**, the support bracket **20** and the display module **10** coupled to the support arm **100** may be moved upward and downward with respect to the support stand **30**.
(29) When the support arm **100** is positioned at the lowermost end of the support stand **30**, the display module **10** arranged in the landscape orientation may not be sufficiently spaced apart from

a floor surface in the upper and lower side direction, and a corner portion of the display module **10** may collide with the floor surface, causing the display module **10** to be damaged. The "floor surface" may refer to a physical space in which the support stand **30** is placed. The "floor surface" may refer to some area on a desk, regardless of its name.
(30) In related devices, in order to prevent collision with the floor surface during a rotation of the display module, the user needs to position the display module at the uppermost end of the support stand and then rotate the display module. However, when the user inadvertently rotates the display module without positioning the display module at the uppermost end of the support stand, the display module may collide with the floor surface, so that the display module is damaged as described above.
(31) According to an embodiment, the display module **10** may be prevented from colliding with the floor surface due to the rotation thereof. Specifically, the display module **10** may be provided to be rotatable with respect to the support devices **20**, **100**, and **30** only when the display module **10** is spaced apart from the floor surface by a distance greater than or equal to a predetermined distance, thereby preventing the display module **10** from colliding with the floor surface.
(32) According to an embodiment, the rotation of the display module **10** may be restricted when the display module **10** is spaced apart from the floor surface by a distance less than the predetermined distance, and is likely to collide with the floor surface due to the rotation thereof. The rotation of the display module **10** may be allowed at a position in which the display module **10** is unlikely to collide with the floor surface even by a rotation thereof. Accordingly, when the display module **10** is in a rotatable state, the display module **10** may not collide with the floor surface.
(33) FIG. **4** is a front exploded perspective view illustrating a display apparatus according to an embodiment. FIG. **5** is a rear exploded perspective view illustrating a display apparatus according to an embodiment.
(34) Referring to FIGS. **4** and **5**, the display apparatus **1** may include the display module **10** and the support devices **20**, **100**, and **30** for supporting the display module **10** to be rotatable and vertically movable.
(35) The display module **10** may include an upper groove **13** and a lower groove **14** provided on a rear surface thereof.
(36) The support devices **20**, **100**, and **30** include the support bracket **20** provided to be coupled to the rear surface of the display module **10**, the support arm **100** coupled to the support bracket **20** and rotatably supporting the support bracket **20**, and the support stand **30** coupled to the support arm **100** and supporting the support arm **100** to enable a upward and downward movement.
(37) The support bracket **20** may be coupled to the rear surface of the display module **10** with an upper protrusion **21***a* and a lower protrusion **22***a*, which will be described below, being inserted into the upper groove **13** and lower groove **14** of the display module **10**, respectively. The support bracket **20** may be coupled to the rear surface of the display module **10** without a separate fastener.
(38) The support bracket **20** may be provided to be coupled to the display module **10** and the support arm **100**. The support bracket **20** may be coupled to the display module **10** with the support arm **100** coupled thereto.
(39) The support arm **100** may rotatably support the support bracket **20**. The support arm **100** may rotatably support the support bracket **20** to thereby rotatably support the display module **10** coupled to the support bracket **20**.
(40) The support stand **30** may be provided to stand on the floor surface by itself. The support stand **30** may include a stand plate **32** provided to be in contact with the floor surface and a stand body **31** coupled to the support arm **100**.
(41) The stand body **31** may include a stand rail **33**. The stand rail **33** may guide an upward and downward movement of the support arm **100** with respect to the stand body **31**.
(42) FIGS. **6** and **7** are views illustrating a support bracket of a display apparatus according to an embodiment.

(43) Hereinafter, the support bracket **20** according to an embodiment will be described in detail with reference to FIGS. **6** and **7**.
(44) Referring to FIGS. **6** and **7**, the support bracket **20** includes the upper protrusion **21***a* and the lower protrusion **22***a* provided to be respectively inserted into the upper groove **13** and the lower groove **14** of the display module **10**. The support bracket **20** may include a moving bracket **22** that is provided to be movable upward and downward, with respect to the upper protrusion **21***a*, and includes the lower protrusion **22***a*. The moving bracket **22** may be provided to be movable upward and downward within a predetermined range, and may be elastically biased downward by an elastic member, such as an elastic band or an elastic cord, or by a spring.
(45) According to an embodiment, the support bracket **20** may include a rotating bracket **21** provided to rotate together with the display module **10** and including the upper protrusion **21***a*, a moving bracket **22** coupled to be movable upward and downward with respect to the rotating bracket **21** and including the lower protrusion **22***a*, a guide bracket **24** coupled to the rotating bracket **21** to rotate together with the rotating bracket **21**, a bracket cover **23** coupled to a rear surface of the rotating bracket **21**, and a fixed bracket **25** coupled to the support arm **100** so as not to rotate together with the display module **10**. In this regard, the rotating bracket **21** may rotate with respect to the fixed bracket **25**.
(46) The rotating bracket **21** may include the upper protrusion **21***a* provided on the upper end thereof. In addition, the rotating bracket **21** may include a guide hole **21***b* into which a second fixing member **27** to be described below is inserted. The guide hole **21***b* may be provided in an arc shape so that the second fixing member **27** may be fixed at a predetermined position even when the rotating bracket **21** rotates.
(47) The moving bracket **22** may be coupled to a lower end of the rotating bracket **21**. The moving bracket **22** may be coupled to the rotating bracket **21** to be movable upward and downward with respect to the rotating bracket **21**. The moving bracket **22** may be elastically biased downward by an elastic member, such as an elastic band or an elastic cord, or by a spring. The moving bracket **22** may include a lower protrusion **22***a* and a coupling hole **22***b* through which a coupling member for coupling the moving bracket **22** and the rotating bracket **21** passes. The coupling member may move upward and downward within the coupling hole **22***b*. The coupling hole **22***b* may limit the range of the upward and downward movement of the coupling member, and by limiting the upward and downward movement range of the coupling member, the upward and downward movement range of the moving bracket **22** may be limited. In addition, the moving bracket **22** may include a rear protrusion **22***c* protruding rearward of the support bracket **20**.
(48) The rear protrusion **22***c* may move upward and downward within a cover hole **23***a* of the bracket cover **23**. Similar to the coupling hole **22***b*, the cover hole **23***a* may limit the range of upward and downward movement of the rear protrusion **22***c*. As the cover hole **23***a* limits the upward and downward movement range of the rear protrusion **22***c*, the upward and downward movement range of the moving bracket **22** may be limited. The upward and downward movement range of the moving bracket **22** may be limited by a hole having a smaller upward and downward movement range between the coupling hole **22***b* and the cover hole **23***a*.
(49) The user may insert the upper protrusion **21***a* into the upper groove **13**, position the lower protrusion **22***a* to correspond to the lower groove **14**, and then grasp the rear protrusion **22***c* to move the rear protrusion **22***c* upward, so that the support bracket **20** is coupled to the rear surface of the display module **10**. As the rear protrusion **22***c* is moved upward, the support bracket **20** may be inserted into a bracket groove provided on the rear surface of the display module **10**. When the user releases the rear protrusion **22***c* with the support bracket **20** being inserted into the bracket groove, the moving bracket **22** may be moved downward by the elastic force of the elastic member, and the lower protrusion **22***a* may be inserted into the lower groove **14**. As the upper protrusion **21***a* is inserted into the upper groove **13** and the lower protrusion **22***a* is inserted into the lower groove **14**, the support bracket **20** may be coupled to the rear surface of the display module **10**.

(50) The guide bracket **24** may rotate together with the rotating bracket **21** and the bracket cover **23**. The guide bracket **24** may rotate together with the display module **10**. The guide bracket **24** may include a locking groove **24***a*. The locking groove **24***a* will be described below. In addition, the guide bracket **24** may further include a link guide portion **24***b*.
(51) The fixed bracket **25** may be provided to not rotate with respect to the support stand **30**, even when the display module **10** rotates. The fixed bracket **25** may be coupled to the support arm **100**. As the fixed bracket **25** is coupled to the support arm **100**, the support bracket **20** and the support arm **100** may be coupled to each other.
(52) A first fixing member **26** and a second fixing member **27** may be provided to couple the rotating bracket **21** and the fixed bracket **25**, respectively. The first fixing member **26** and the second fixing member **27** may be provided to pass through the rotating bracket **21** and the fixed bracket **25**, respectively.
(53) The first fixing member **26** may pass through a hole provided in the center of the rotating bracket **21**. An end portion of the first fixing member **26** having passed through the rotating bracket **21** may be coupled to a first fixing head **26***a*. The first fixing head **26***a* may have a diameter larger than that of the hole so as not to pass through the hole of the rotating bracket **21**.
(54) The second fixing member **27** may pass through a guide hole **21***b* of the rotating bracket **21** and a through hole of the fixed bracket **25**. An end portion of the second fixing member **27** protruding to the rear side of the support bracket **20** by passing through the guide hole **21***b* and the through hole may be coupled to a second fixing head **27***a*. The second fixing head **27***a* may have a diameter larger than the width of the guide hole **21***b* and the size of the through hole so as not to pass through the guide hole **21***b* and the through hole.
(55) The second fixing member **27** may be located in the guide hole **21***b* of the rotating bracket **21**. Because the guide hole **21***b* is provided in an arc shape, even when the second fixing member **27** that does not rotate together with the display module **10** is located in the guide hole **21***b*, the rotating bracket **21** may rotate together the display module **10**. In this regard, because the guide hole **21***b* is provided in an arc-shape, the second fixing member **27** may not restrict the rotation of the rotating bracket **21**.
(56) FIG. **8** is a view illustrating an internal structure of a support arm when a pivot rotation of a display module is restricted in a display apparatus according to an embodiment. FIG. **9** is a view illustrating an internal structure of a support arm when a pivot rotation of a display module is allowed in a display apparatus according to an embodiment. Hereinafter, the support arm **100** according to the embodiment will be described in detail with reference to FIGS. **8** and **9**.
(57) Referring to FIGS. **8** and **9**, the support arm **100** may include a first bracket **101** and a second bracket **102** provided to be coupled to the fixed bracket **25**. The first bracket **101** and the second bracket **102** may be coupled to the fixed bracket **25** by a fastening member. As the first bracket **101** and the second bracket **102** are coupled to the fixed bracket **25**, the support arm **100** and the support bracket **20** may be coupled to each other.
(58) The support arm **100** may include a first fixing link **110** and a second fixing link **120** that are coupled to the support stand **30** to be movable in the upward and downward direction. The first fixing link **110** and the second fixing link **120** may be laterally arranged side by side. The first fixing link **110** and the second fixing link **120** may be coupled to each other by a fastening member.
(59) The first fixing link **110** may include a first rail inserting protrusion **111** that is inserted into the stand rail **33** of the support stand **30** to move upward and downward along the stand rail **33**.
(60) The second fixing link **120** may include a second rail inserting protrusion **121** that is inserted into the stand rail **33** of the support stand **30** to move upward and downward along the stand rail **33**.
(61) The support arm **100** may include a first spring **103** disposed between the first fixing link **110** and the first bracket **101** and a second spring **104** disposed between the second bracket **102** and the second fixing link **120**. The first spring **103** and the second spring **104** may include torsion springs.

(62) The support arm **100** may support the support bracket **20** to be rotatably movable within a predetermined range with respect to a rotation axis extending in a leftward and rightward direction. Such a rotational movement is referred to as a tilt. The support arm **100** may support the support bracket **20** to enable a tilt operation, thereby supporting the display module **10** coupled to the support bracket **20** to enable a tilt operation.
(63) The first spring **103** and the second spring **104** may provide an elastic force such that the display module **10** may be maintained in a fixed state at any position within a tilting range of the display module **10**.
(64) The support arm **100** may include a locking link **130** provided to move forward and backward with respect to the first fixing link **110**, and a rotating link **140** provided to rotate such that the locking link **130** is moved. The rotating link **140** may also be provided to be rotated by a movement of the locking link **130**.
(65) The support bracket **20** may include the locking groove **24***a* provided on the rear surface thereof. Specifically, the support bracket **20** may include the guide bracket **24** including the locking groove **24***a*. The guide bracket **24** may be provided to rotate together with the display module **10** during rotation of the display module **10**.
(66) According to an embodiment, the locking groove **24***a* may include three locking grooves to allow for rotation of the support bracket **20** in both directions. The three locking grooves **24***a* may be spaced apart from each other to correspond with the locking link **130**, which will be described below, whenever the support bracket **20** rotates at an interval of 90 degrees. However, embodiments are not restricted thereto, and the locking groove **24***a* may be provided in one or two grooves.
(67) Referring to FIG. **8**, the locking link **130** may be inserted into the locking groove **24***a*. When the locking link **130** is inserted into the locking groove **24***a* of the support bracket **20**, the rotation of the support bracket **20** may be restricted. In this regard, upon the locking link **130** of the support arm **100** being inserted into the locking groove **24***a* of the support bracket **20**, the rotation of the support bracket **20** and the display module **10** coupled to the support bracket **20** may be restricted.
(68) Referring to FIG. **9**, the locking link **130** may be withdrawn from the locking groove **24***a* by moving backward. When the locking link **130** is withdrawn from the locking groove **24***a* of the support bracket **20**, the rotation of the support bracket **20** may be allowed. In this regard, upon the locking link **130** of the support arm **100** being withdrawn from the locking groove **24***a* of the support bracket **20**, the rotation of the support bracket **20** and the display module **10** coupled to the support bracket **20** may be allowed. When the locking link **130** is withdrawn from the locking groove **24***a*, the rotation restriction of the support bracket **20** may be released.
(69) FIG. **10** is a cross-sectional view taken along line A-A′ of FIG. **8**. FIG. **11** is a cross-sectional view taken along line B-B′ of FIG. **9**.
(70) Hereinafter, a locking mechanism of the support arm **100** and the support bracket **20** according to an embodiment will be described in detail with reference to FIGS. **10** and **11**.
(71) Referring to FIG. **10**, in a state in which the support arm **100** is not positioned at the maximum height with respect to the support stand **30**, the locking link **130** may be inserted into the locking groove **24***a*.
(72) The locking link **130** may include a locking portion **131** inserted into or withdrawn from the locking groove **24***a*, an extension portion **132** extending rearward of the locking portion **131**, and an insertion groove **133** formed in the extension portion **132**.
(73) The first fixing link **110** may include a guide groove **112** for guiding the forward and backward movement of the locking link **130**. The locking link **130** may move forward and backward within the guide groove **112**.
(74) The support arm **100** may include an elastic member **134**, such as an elastic band or an elastic cord, or by a spring, for providing an elastic force such that the locking link **130** is moved forward with respect to the first fixing link **110**. The elastic member **134** may elastically bias the locking

link **130** to be moved forward. For the locking link **130** to be moved forward (e.g., away from the support stand **30** and towards the support bracket **20**) by the elastic force of the elastic member **134**, one end **134***a* of the elastic member **134** may be connected to the locking link **130**, and the other end **134***b* of the elastic member **134** may be connected to the first fixing link **110**.

(75) According to an embodiment, upon the locking portion **131** being inserted into the locking groove **24***a*, the distance between the one end **134***a* of the elastic member **134** and the other end **134***b* of the elastic member **134** may have a minimum distance. Upon the locking portion **131** being withdrawn from the locking groove **24***a*, the distance between the one end **134***a* of the elastic member **134** and the other end **134***b* of the elastic member **134** may be increased. The elastic member **134** may provide an elastic force such that the distance between the one end **134***a* and the other end **134***b* is shortened. The elastic member **134** may include a tension spring.

(76) The rotating link **140** may include an inserting protrusion **141** provided to be inserted into the insertion groove **133** of the locking link **130** and an extension protrusion **142** extending to the inner side of the support stand **30**. The inserting protrusion **141** and the extension protrusion **142** may be integrally formed with each other to rotate together. The inserting protrusion **141** and the extension protrusion **142** may extend in different directions. The inserting protrusion **141** may be provided shorter than the extension protrusion **142**.

(77) The support stand **30** may include a rotation guide portion **34** provided on the inner upper side thereof. The rotation guide portion **34** may be provided to guide the rotation of the rotating link **140**. The rotation guide portion **34** may be disposed adjacent to the upper end of the support stand **30**. The rotation guide portion **34** may include a rotation guide surface **34***a* provided to be in contact with the rotating link **140**. The rotation guide surface **34***a* may be formed to be oblique at a predetermined angle with respect to the horizontal direction.

(78) Referring to FIG. **10**, in a state in which the support arm **100** is not positioned at the uppermost end of the support stand **30**, the rotating link **140** may not be in contact with the rotation guide portion **34** of the support stand **30**. In addition, the locking portion **131** of the locking link **130** may be inserted into the locking groove **24***a*. The support arm **100** may be movable downward with the locking portion **131** being inserted into the locking groove **24***a*.

(79) Referring to FIG. **11**, in a state in which the support arm **100** is positioned at the uppermost end of the support stand **30**, the rotating link **140** may be in contact with the rotation guide portion **34** of the support stand **30**. In addition, the locking portion **131** of the locking link **130** may be withdrawn from the locking groove **24***a*. Because the support arm **100** is positioned at the uppermost end of the support stand **30**, upward movement of the support arm **100** may be restricted.

(80) Referring to FIGS. **10** and **11**, when the support arm **100** moves upward to the uppermost end of the support stand **30**, the extension protrusion **142** of the rotating link **140** contacts the rotation guide surface **34***a* of the support stand **30**. When the support arm **100** moves upward while the extension protrusion **142** is in contact with the rotation guide surface **34***a*, the extension protrusion **142** is rotated in a first direction by the rotation guide surface **34***a*. The first direction may indicate a clockwise direction with reference to FIGS. **10** and **11**.

(81) The extension protrusion **142** may be rotated until the extension protrusion **142** is arranged in parallel to the rotation guide surface **34***a*. In this regard, the extension protrusion **142** may be rotated until a first angle at which the extension protrusion **142** is inclined with respect to the horizontal direction is equal to a second angle at which the rotation guide surface **34***a* is inclined with respect to the horizontal direction.

(82) When the extension protrusion **142** is rotated a predetermined angle by the rotation guide surface **34***a*, the inserting protrusion **141** may be rotated the predetermined angle together with the extension protrusion **142**. When the inserting protrusion **141**, while being inserted into the insertion groove **133**, rotates in the first direction, the locking link **130** may be moved backward (e.g., toward the support stand **30** and away from the support bracket **20**). That is, when the rotating link

140 rotates in the first direction, the locking link **130** may be moved backward. When the rotational force of the rotating link **140** is greater than the elastic force of the elastic member **134**, the locking link **130** may be moved backward.

(83) When the locking link **130** is moved backward, the locking portion **131** may be withdrawn from the locking groove **24***a*. When the locking portion **131** is withdrawn from the locking groove **24***a*, the rotation of the support bracket **20** with respect to the support arm **100** is not restricted.

(84) According to an embodiment, when the support arm **100** is positioned at the uppermost end of the support stand **30**, the locking link **130** may be moved backward and the rotation of the support bracket **20** may be allowed. When the rotation of the support bracket **20** is allowed, the rotation of the display module **10** coupled to the support bracket **20** may be allowed.

(85) When the display module **10** is positioned at the maximum height of the support stand **30**, the display module **10** may not collide with the floor surface even with a rotation of the display module **10** arranged in a landscape orientation. Because the display module **10** is sufficiently spaced apart from the floor surface, the display module **10** may not collide with the floor surface even when the display module **10** rotates.

(86) When the display module **10** is moved downward from the maximum height of the support stand **30**, the locking link **130** may be moved forward by the elastic force of the elastic member **134** and thus inserted into the locking groove **24***a*. Together with the forward movement of the locking link **130**, the insertion groove **133** is moved forward to rotate the inserting protrusion **141** in a second direction opposite to the first direction. When the support arm **100** moves downward, the rotating link **140** may be separated from the rotation guide portion **34**.

(87) According to one or more embodiments, a display apparatus capable of preventing a display module from colliding with a floor surface due to being pivotally rotated can be provided.

(88) According to one or more embodiments, a display apparatus in which a display module is provided to perform a pivot rotation with respect to a support device only when the display module is spaced apart from the floor surface by a predetermined distance or more can be provided.

(89) According to one or more embodiments, a display apparatus in which rotation of a display module is restricted at a position having a possibility of colliding with a floor surface due to being rotated can be provided.

(90) Although aspects of embodiments have been shown and described, it will be appreciated by those skilled in the art that changes and modifications may be made without departing from the principles and scope of the disclosure, the scope of which is defined in the claims and their equivalents.

Claims

1. A display apparatus comprising: a display panel; a support bracket coupled to a rear surface of the display panel and including a locking groove, the support bracket being configured to rotate between a first orientation in which a long side of the display panel is laterally disposed and a second orientation in which the long side of the display panel is longitudinally disposed; a support arm coupled to the support bracket and supporting the display panel; and a support stand supporting the support arm and configured to allow the support arm to move along a vertical direction between a first position and a second position lower than the first position, wherein the support arm comprises: a rotating link configured to rotate in a first direction as the support arm moves from the second position to the first position, wherein the rotating link comprises a first protrusion; and a locking link configured to move from a locked position in which the locking link is inserted into the locking groove to restrict rotation of the display panel, to an unlocked position in which the locking link is withdrawn from the locking groove to allow rotation of the display panel, based on the rotating link rotating in the first direction, wherein the locking link defines an insertion groove which accommodates the first protrusion and moves along a horizontal direction

as the locking link moves from the locked position to the unlocked position.
2. The display apparatus of claim 1, wherein the support arm further comprises a fixing link coupled to the support stand, and wherein the insertion groove moves along the horizontal direction with respect to the fixing link.
3. The display apparatus of claim 2, wherein the locking link is further configured to be inserted into the locking groove by moving toward the support bracket, and to be withdrawn from the locking groove by moving away from the support bracket.
4. The display apparatus of claim 2, wherein the support arm further comprises an elastic member configured to bias the locking link toward the support bracket.
5. The display apparatus of claim 4, wherein a first end of the elastic member is connected to the fixing link, and a second end of the elastic member is connected to the locking link.
6. The display apparatus of claim 4, wherein the locking link is further configured to move to the locked position based on an elastic force of the elastic member when the support arm moves from the first position to the second position.
7. The display apparatus of claim 6, wherein the rotating link is further configured to rotate in a second direction opposite to the first direction based on the locking link moving from the unlocked position to the locked position.
8. The display apparatus of claim 4, wherein the locking link comprises: a locking portion configured to extend be inserted into the locking groove or withdrawn from the locking groove; and an extension portion extending toward the support stand and including the insertion groove.
9. The display apparatus of claim 8, wherein the rotating link further comprises a second protrusion extending into the support stand and configured to rotate together with the first protrusion.
10. The display apparatus of claim 9, wherein the support stand comprises a rotation guide portion configured to contact the second protrusion and rotate the rotating link in the first direction as the support arm moves from the second position to the first position.
11. The display apparatus of claim 10, wherein the rotation guide portion is provided adjacent to an upper end of the support stand.
12. The display apparatus of claim 10, wherein the rotation guide portion comprises a rotation guide surface provided at an angle oblique to the horizontal direction, and wherein the rotation guide surface is configured to guide the second protrusion such that the second protrusion rotates in the first direction to be arranged in parallel with the rotation guide surface.
13. The display apparatus of claim 1, wherein the support bracket is detachably coupled to the rear surface of the display panel.
14. A display stand comprising: a support bracket having a locking groove formed therein, and configured to support a display panel and rotate between a first orientation and a second orientation; a support arm configured to support the support bracket; and a support stand configured to slidably support the support arm between a first position and a second position lower than the first position along a vertical direction, wherein the support arm comprises: a rotating link configured to rotate in a first direction as the support arm moves from the second position to the first position, wherein the rotating link comprises a first protrusion; and a locking link configured to move from a locked position in which the locking link is inserted into the locking groove to restrict rotation of the display panel, to an unlocked position in which the locking link is withdrawn from the locking groove to allow rotation of the display panel, based on the rotating link rotating in the first direction, wherein the locking link defines an insertion groove which accommodates the first protrusion and moves along a horizontal direction as the locking link moves from the locked position to the unlocked position.
15. The display stand of claim 14, wherein the support arm is configured to move in the vertical direction between the first position and the second position, and wherein the horizontal direction is perpendicular to the vertical direction.
16. The display stand of claim 15, wherein the locking groove is aligned with the locking link

based on the support bracket being in the first orientation, and wherein another locking groove is formed in the support bracket and is aligned with the locking link based on the support bracket being in the second orientation.

17. The display stand of claim 16, wherein the support arm further comprises a fixed portion, and wherein the locking link is further configured to slide along the fixed portion in the horizontal direction.

18. The display stand of claim 17, further comprising a spring connected between the fixed portion and the locking link, and configured to bias the locking link to the locked position.

19. A method of limiting rotation of a support bracket that is slidable along a support stand between a first position and a second position lower than the first position along a vertical direction, the method comprising: restricting rotation of the support bracket based on the support bracket being at the second position; sliding the support bracket from the second position to the first position; rotating a rotating link in a first direction as the support bracket moves to the first position, wherein the rotating link comprises a first protrusion; withdrawing a locking link from a locking groove formed in the support bracket based on the support bracket sliding from the second position to the first position, wherein the locking link defines an insertion groove which accommodates the first protrusion and moves along a horizontal direction as the locking link is withdrawn from the locking groove.

20. The method of claim 19, wherein the rotating the rotating link in the first direction moves the locking link.